

Course “Simulation” - Exam Project, 2025/2026

Student: XXXXXXXXXX

Project Title: Glassworks Simulation Model and Analysis

Introduction

This project consists of a simulation model for the production flow of a glass windows manufacturer in order to understand key issues and help in the decisions related to the scale of the operation.

The main goals are to understand where bottlenecks happen and help reduce their impact, finding the minimum resources required to keep a satisfying flow of production.

Model

The model is described as follows. The manufacturer orders raw glass plates every specified number of days (**RestockRate** and **RestockQuantity**) to produce the processed glass used in commercial and residential windows.

Since orders arrival may depend on seasonal and cyclical components other than major events, the real distribution of orders can't be known without actual data. The lack of this information requires us to approximate the process as independent events that arrives at exponential rate (**OrderRate**), a Poisson process.

Each order has few parameters to better capture the variability of the manufacturing process:

- The cut windows **width**, **height** and **aspectRatio** depend on the order type, the manufacturer produces residential windows with a width ranging from **600mm to 1200mm** and commercial ones ranging from **400mm to 3100mm** with aspect ratios ranging from **1:1 to 2:1**. Since we have no information on the distribution we assume them to follow a triangular distribution centered in the middle of those ranges.
- Products can have shapes more complex than a simple rectangle, the manufacturer saw that shapes complexity follows this distribution: simple rectangles constitute around **80%** of the orders, more complex shapes like circles and ovals **15%** and then custom shapes are around **5%**. This has been modeled as the **shapeComplexity** coefficient with values **1.0**, **1.25** and **1.5**.
- Windows of both residential and commercial structures can be reinforced with an interlayer material which is required in some countries for glass railings and other

uses. The manufacturer knows that roughly **15%** of his windows have to go through this treatment, which is a long process.

- The raw glass plates have to be cut in a determined **quantity** of pieces, this is modelled by an uniform discrete random variable between **1 and 20**, the raw plates are different correct sizes for the process.

The raw glass material then goes through different machines:

1. **Cutter**: cuts the material in different pieces.
2. **Edger**: smooths borders for handling safety.
3. **Washer**: cleans the product of impurities.
4. **Press**: inserts the interlayer reinforcement into the plates lightly and removes air bubbles. Only needed for reinforced windows.
5. **Autoclave** finalizes the process and bonds the reinforcement layer to the glass plates. Only needed for reinforced windows.

We assume the order to be already optimized for a single glass plate (**GlassPiece**), we consider one order to be a single shape. Cut pieces are then treated separately unless they require passing through the autoclave which works in batches.

The timings of each machine are calculated using both deterministic formulas (that take in input random variables generated during the order) to try to model their real operations approximatively:

1. **Cutter**: $triangular(20, 35, 60) + \left(2 \times \frac{width+height}{1000}\right) \times agentShapeComplexity \times 10$ seconds.
2. **Edger**: $\frac{2 \times (width+height)}{1000} \times 30 \times shapeComplexity$ seconds.
3. **Washer**: $triangular(60, 90, 150)$ seconds.
4. **Press**: Random $triangular(5, 10, 18)$ minutes.
5. **Autoclave**: Random $triangular(120, 180, 240)$ minutes (for **BatchSize** pieces).

For every completed batch of final products (**ShippingSize**) a transport vehicle is called to proceed with the shipping (this is a simplification and could split the same order in different shippings).

Every piece is moved by free moving agents representing workers that transport glass using carts or suction cups, as using forklifts would have high chances of breaking the plates.

Implementation

The model for this project has been implemented in Anylogic as follows.

At a fixed rate a truck arrives and ships a defined quantity of **Raw Material** to be stock in the storage.

Orders get generated from a source block with an exponential interarrival time and get assigned the random values as described previously, those Orders then pass their parameters to the **Raw Material** agents stocked in the storage using a **Match** block in order to retrieve them and start processing, transferring the required parameters to the piece to be treated.

The orders then get sorted in two wait queues, one for reinforced windows orders and one for other ones. Those two queues use **Wait** blocks to achieve blocked queues. When a piece exits the last machine process and it's ready to be shipped the first order in the corresponding queue decreases its quantity, if it reaches zero it's removed from the queue by freeing the specific agent (avoiding timing issues). This is used in conjunction of **TimeMeasureStart** and **TimeMeasureEnd** blocks to estimate **lead times** (without shipping).

For transporting custom **Transporter** agents are seized from a single pool to move pieces to the machines and out of them.

The material waits in a queue to seize a free cutting machine from the **cutterPool** resource pool, then it gets released by the transporter and put in the seized machine, which then waits for a delay time to process, then gets released with a **Release** block.

The output of the machine is sent to a **Split** node to represent the creation of different cut pieces and moves to the next processing step where they get their edges smooth to avoid injuries, this is done singularly for each piece. This process is modeled in the same way as the cutter with **Queue**, **Seize**, **Delay** and **Release** nodes.

Processed pieces then have to get washed, the material goes through the same set of blocks, this usually involves rolling machines so we can consider a single piece to be treated at once here too.

Then if the plate doesn't require to be reinforced it's sent directly to the **queueProduct** shipping queue, otherwise it's sent to the **Press** and the **Autoclave**. Before getting to the autoclave, a parametrical amount of pieces is batched using a **Batch** block to process more items at the same time, this amount has to be tuned to avoid bottlenecks and large lead times.

After the last machine operation the plates are queued in the shipping output dock, when a **ShippingSize** number of pieces is done they get into a vehicle agent to be shipped.

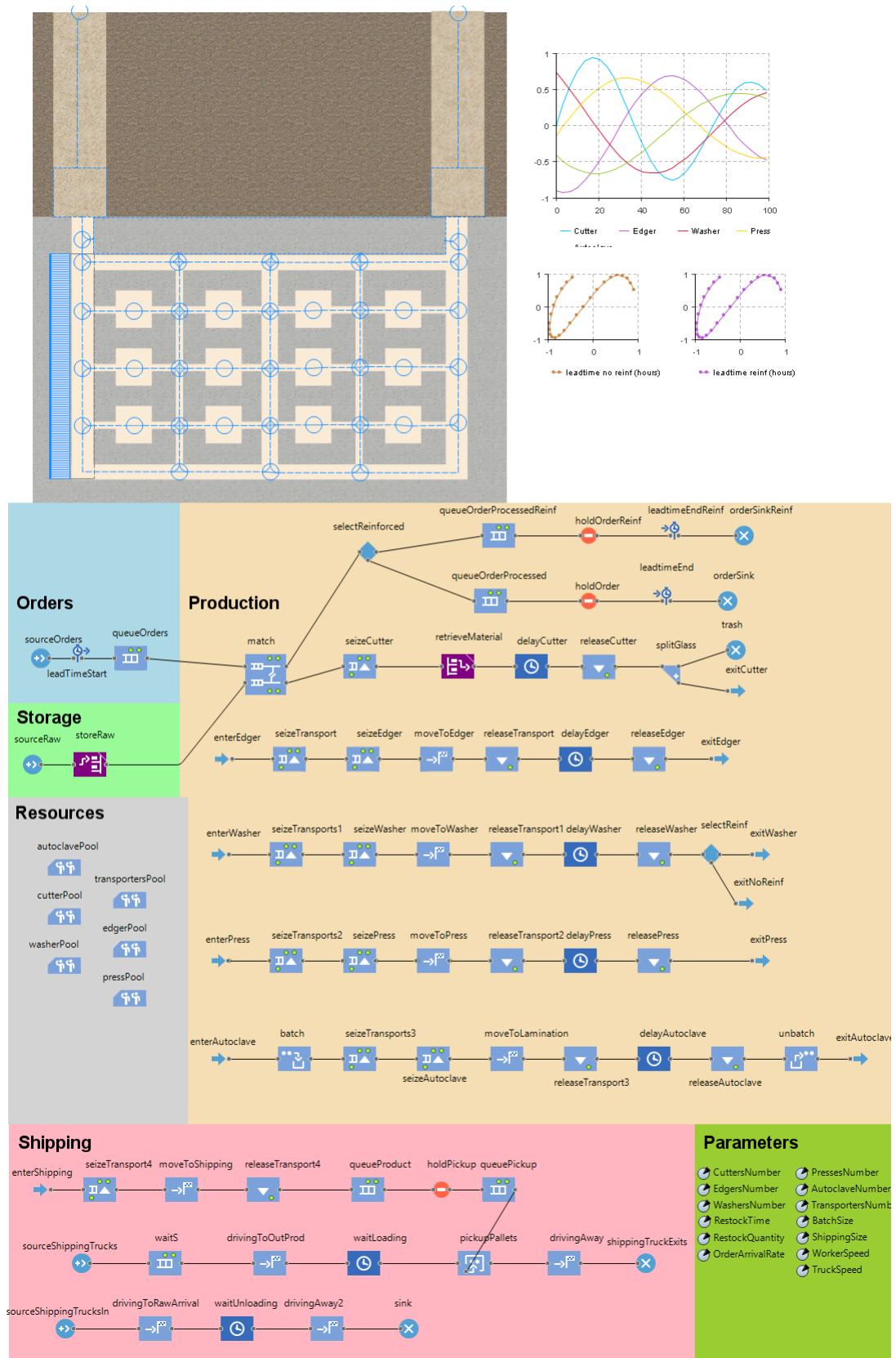


Figure 1: The model as presented inside Anylogic's Main agent visualization.

Key System Parameters (KSP) and Key Performance Indicators (KPI)

The simulation output depends on different parameters that can be controlled by the manufacturer, tweaking those can achieve varying outcomes.

The parameters considered for each simulation are:

- **Quantity of each machine:** a maximum amount for each machine, it's essential to find bottlenecks and distribute the flow of production.
- **Transporters number:** indicates the number of workers for transporting and working on machines, the goal is to have the least amount of workers on idle during the day but still be able to process all the orders continuously.
- **Restock Rate:** the rate at which the raw material storage is restocked to satisfy the arrival of the orders.
- **Restock Quantity:** the quantity to be ordered every time the storage needs to be restocked.
- **Batch Size:** the amount of pieces to be batched for the longest process that is the Autoclave press.
- **Shipping Size:** the finished products are shipped in batches, this is tweaked to ensure a continuous output flow without big queues in the shipping dock.

Other parameters were added to simplify the experiments, they are not meant to be modified by manufacturer decisions: **Workers speed**, **Truck Speed** and **OrderArrivalRate**.

The **Key Performance Indicators** to be considered are:

- **Machine pool utilization:** indicates how much time is the single machines pool utilized instead of being idle. This is used to figure out the optimal amount of machines for each production step.
- **Lead times (divided by order type):** helps understanding if the processes flow can keep up with the orders on the long run.
- **Transporter pool utilization:** to check if workers do not have high idle times.
- **Total Throughput:** the total amount of batches delivered in the time window of the simulation.

Experimental analysis

The next experiments try to find and understand how tweaking the parameters influence our indicators, each experiment considers **160 hours** (around one month) and in total **100 replications** are considered, the mean value of each indicator is collected.

First Experiment

A starting configuration has been intuitively considered as follows:

Table 1: Experiment Parameters

Parameter	Value
EdgersNumber	3
CuttersNumber	2
WashersNumber	2
AutoclaveNumber	2
PressesNumber	3
TransportersNumber	20
BatchSize	20
RestockRate	1/day
RestockQuantity	120
OrderArrivalRate	120/day
ShippingSize	50
WorkerSpeed	1.5 m/s
TruckSpeed	2.8 m/s

Table 2: Results (95% Confidence Interval)

Indicator	Mean $\pm$ Interval
cutterUtilization	0.1858 ± 0.0128
edgerUtilization	0.2956 ± 0.0041
washerUtilization	0.7180 ± 0.0137
pressUtilization	0.5762 ± 0.0141
autoclaveUtilization	0.4243 ± 0.0124
transportersUtilization	0.9743 ± 0.0045
leadTime	10.18 ± 0.87 h
leadTimeReinf	23.18 ± 1.42 h
throughput	6553.5 ± 177.1
queueShippingSize	23.24 ± 2.30

From this first experiments we can look at the utilization of each machine to adapt the system to the demand of 120 **Orders** per day. Interpreting the data we can see that some machines are underutilized while some are overutilized, a new experiment can be done considering one more **Washer** and reduce both **Cutters** and **Edgers** by one.

Second Experiment

Table 3: Experiment Parameters

Parameter	Value
EdgersNumber	2
CuttersNumber	1
WashersNumber	3
AutoclaveNumber	2
PressesNumber	3
TransportersNumber	20
BatchSize	20
RestockRate	1/day
RestockQuantity	120
OrderArrivalRate	120/day
ShippingSize	50
WorkerSpeed	1.5 m/s
TruckSpeed	2.8 m/s

Table 4: Results (95% Confidence Interval)

Indicator	Mean $\pm$ Interval
cutterUtilization	0.3115 ± 0.0120
edgerUtilization	0.4737 ± 0.0051
washerUtilization	0.5089 ± 0.0081
pressUtilization	0.6587 ± 0.0150
autoclaveUtilization	0.5050 ± 0.0131
transportersUtilization	0.8815 ± 0.0140
leadTime	5.33 ± 0.83 h
leadTimeReinf	15.01 ± 1.38 h
throughput	7439.0 ± 165.6
queueShippingSize	24.28 ± 2.29

In this experiment the machines utilization has been increased, we can also say probabilistically that both the expected throughput and lead times would decrease as their

confidence intervals don't intersect. Also the 88% transporters utilization means that the amount of workers can be reduced.

Third Experiment

Table 5: Experiment Parameters

Parameter	Value
EdgersNumber	2
CuttersNumber	1
WashersNumber	3
AutoclaveNumber	2
PressesNumber	3
TransportersNumber	16
BatchSize	20
RestockRate	1/day
RestockQuantity	120
OrderArrivalRate	120/day
ShippingSize	50
WorkerSpeed	1.5 m/s
TruckSpeed	2.8 m/s

Table 6: Results (95% Confidence Interval)

Indicator	Mean $\pm$ Interval
cutterUtilization	0.3508 ± 0.0152
edgerUtilization	0.4565 ± 0.0075
washerUtilization	0.4810 ± 0.0112
pressUtilization	0.6097 ± 0.0183
autoclaveUtilization	0.4566 ± 0.0163
transportersUtilization	0.9449 ± 0.0072
leadTime	8.52 ± 0.94 h
leadTimeReinf	20.11 ± 1.48 h
throughput	6913.5 ± 249.7
queueShippingSize	25.16 ± 2.39

This change has interesting outcomes, the transporters are now around 95% mean utilization, however this has increased the lead times for both types of products. Decisions on tweaking this value depends on external factors, for example the decrease in cost that comes from reducing workers might be worth the extra waiting time before shipping and the decreased throughput, but also the varying amount of work has to be considered.

Fourth Experiment

Here is now considered a higher workload to see how the current system behaves in high demand periods.

Table 7: Experiment Parameters

Parameter	Value
EdgersNumber	2
CuttersNumber	1
WashersNumber	3
AutoclaveNumber	2
PressesNumber	3
TransportersNumber	16
BatchSize	20
RestockRate	1/day
RestockQuantity	150
OrderArrivalRate	150/day
ShippingSize	50
WorkerSpeed	1.5 m/s
TruckSpeed	2.8 m/s

Table 8: Results (95% Confidence Interval)

Indicator	Mean $\pm$ Interval
cutterUtilization	0.5511 ± 0.0156
edgerUtilization	0.4769 ± 0.0084
washerUtilization	0.4167 ± 0.0110
pressUtilization	0.4447 ± 0.0147
autoclaveUtilization	0.2738 ± 0.0116
transportersUtilization	0.9976 ± 0.0001
leadTime	31.29 ± 0.97 h
leadTimeReinf	55.19 ± 1.43 h
throughput	4808.5 ± 171.3
queueShippingSize	25.78 ± 2.25

The results indicate that the transporters cannot keep up with the moving tasks being always at around 100% utilization and the lead times increased reaching over 30 hours. So the experiment is repeated with more workforce until an acceptable flow is reached.

Table 9: Experiment Parameters

Parameter	Value
EdgersNumber	2
CuttersNumber	1
WashersNumber	3
AutoclaveNumber	2
PressesNumber	3
TransportersNumber	30
BatchSize	20
RestockRate	1/day
RestockQuantity	150
OrderArrivalRate	150/day
ShippingSize	50
WorkerSpeed	1.5 m/s
TruckSpeed	2.8 m/s

Table 10: Results (95% Confidence Interval)

Indicator	Mean $\pm$ Interval
cutterUtilization	0.4551 ± 0.0143
edgerUtilization	0.5371 ± 0.0076
washerUtilization	0.5296 ± 0.0117
pressUtilization	0.6295 ± 0.0176
autoclaveUtilization	0.4390 ± 0.0151
transportersUtilization	0.9949 ± 0.0015
leadTime	18.04 ± 1.01 h
leadTimeReinf	36.36 ± 1.54 h
throughput	6918.5 ± 210.2
queueShippingSize	23.77 ± 2.29

With 30 workers the lead times get reduced to around 18 hours, this shows that with increased demand, the amount of workforce has to scale very fast, the manufacturer might have to reconsider using manual transporters for glass or consider splitting the shop in more warehouses.

Conclusions

The experiments show that the machines are more efficient for the specific tasks than intuitively expected and that decreasing the amount of workers might help reduce costs but has an impact resulting in longer expected lead times that scale with the amount of orders. Decisions from the manufacturer might change depending on expected demand, and more tests would be needed to understand specific situations.